

VIET NAM NATIONAL UNIVERSITY HO CHI MINH CITY  
**UNIVERSITY OF ECONOMICS AND LAW**

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FACULTY OF INFORMATION SYSTEMS



# **Causal AI Models: An Automated Treatments Approach to Customer Retention in the Banking Sector**

**Presentation**

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*Presented by:* **Vo Minh Thanh**  
**K214162152**

April – 2025

# Presentation Outline

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- Conclusion

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# I. INTRODUCTION & BACKGROUND

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# Background and Context

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Acquiring a new customer can be 5–25 times more expensive than retaining an existing one.

1

Even a 5% increase in customer retention can boost profits by 25–95%.

2

Rapid innovation from fintechs and digital banks makes switching easier for customers.

3

Lowering churn saves marketing budgets, strengthens market share, and increases ROI.

4

**Banks must adopt data-driven, proactive retention strategies to protect revenue and remain competitive.**

# Problem Statement & Literature Review

## ➔ Topic #1

Lack of Integration of Predictive Modeling and Segmentation

## ➔ Topic #2

Limited Studies on Machine Learning + Causal AI

## ➔ Topic #3

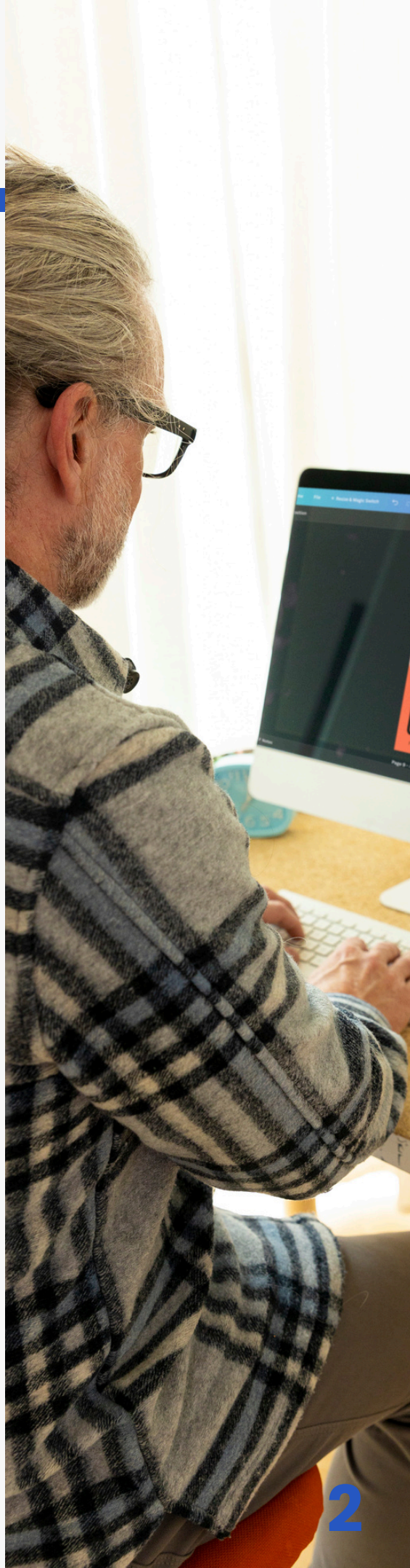
Shortcomings of Traditional Predictive Approaches

## ➔ Topic #4

Single vs. Multi-Treatment Strategies

### Why Causal AI & Multi-Treatment Analysis?

- Causal AI uncovers which interventions truly affect churn (vs. mere correlation), enabling more precise targeting and reducing wasted effort.
- Multi-treatment analysis chooses the best possible intervention among multiple options for each customer segment, leading to significantly better retention outcomes.





# II. RESEARCH OBJECTIVES & METHODOLOGY

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# Ultimate Goal & Key Focus Areas —————

**Reduce customer churn by designing data-driven framework and personalized interventions.**

## **Key Focus Areas**

1. **Predictive Analytics** – Identify at-risk customers.
2. **Customer Segmentation** – Group similar customers for targeted actions.
3. Proposing Multiple Retention **Treatments**
4. **Causal Inference** – Determine which interventions truly reduce churn.
5. Multi-Treatment Analysis – **Optimize** among multiple retention strategies for each customer.

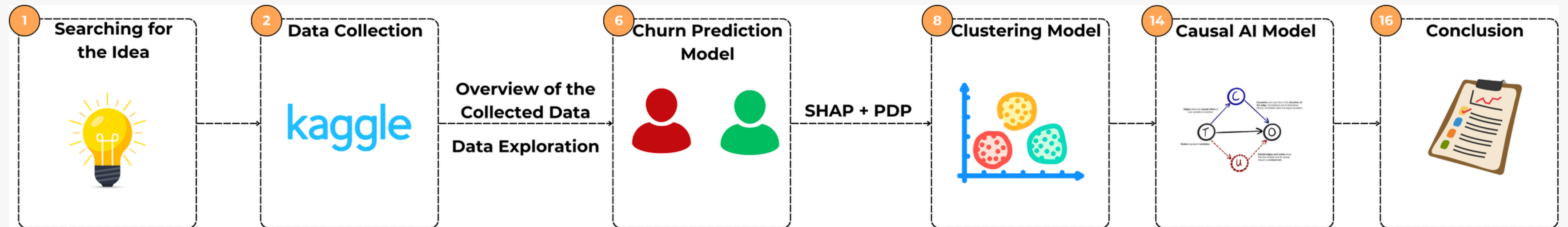
# Causal Questions

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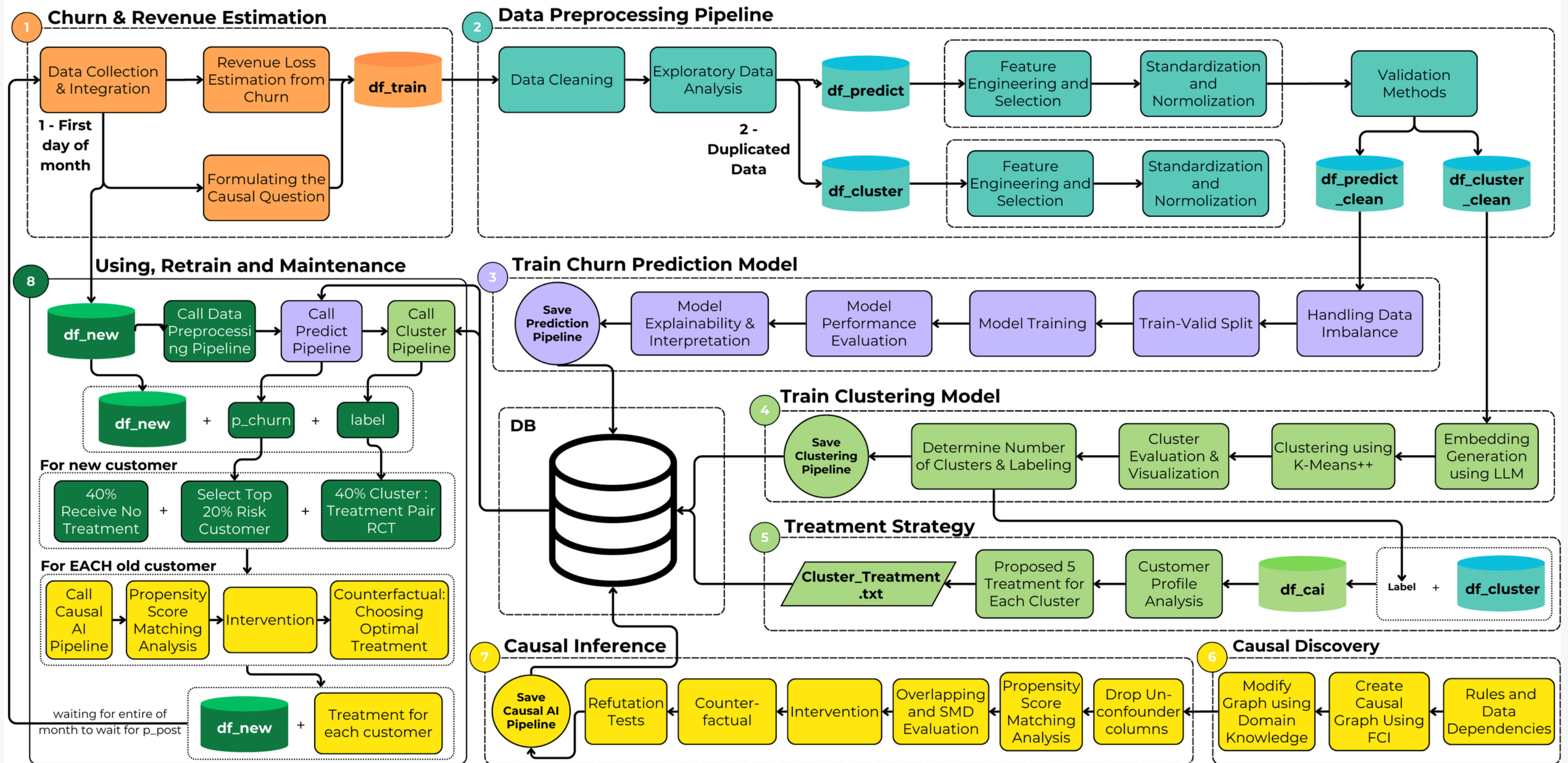
- **Why** do customers leave? (Beyond correlation to causality)
- **Which** intervention effectively reduces churn, and for whom?
- **Counterfactual Analysis:** “What if we had offered a different incentive?”
- **Impact Assessment:** “How much churn reduction can we realistically achieve?”



# Methodology



**Fig 1** – High Level of Work Flow



**Fig 2 - Detail Steps of Methodology**



# Data & Preprocessing

## ➔ 165034 x 15

The dataset used in this study was sourced from Kaggle

- **Cleaning & Transformation:** Handling outliers, missing values, feature engineering
- **Imbalanced Classes:** Techniques like SMOTE to address churn vs. non-churn imbalance.

## ➔ Overview

- **Age:**

Avg = 38 | SD = 8.86 | Range: 18–92 → Diverse age distribution

- **Credit Score:**

Avg = 656 | Range: 350–850 → Typical credit profile

- **Balance:**

Avg = 55,478 ➤ 25% have zero balance → Signs of low engagement

- **Products per Customer:**

Avg = 1.55 → Most hold 1–2 products

- **Credit Card Ownership:**

75% have a credit card

- **Active Members:**

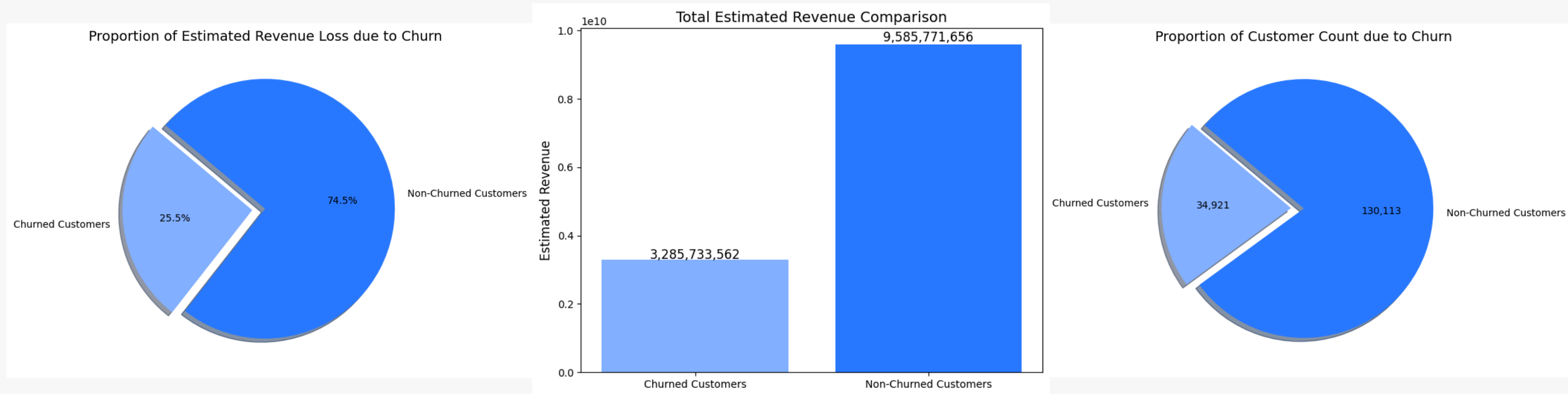
Only ~50% are active → Key churn signal

- **Estimated Salary:**

Avg = 112,547 | High variability → Broad income range



# Why Churn Matters



## Why Churn Matters

- **21.2%** of customers churned → Caused a **23.1%** revenue loss (**~\$6.1B**)
- Loss includes high-value clients → Major impact on cross-sell potential
- Highlights the urgent need for innovative retention strategies
- Churn threatens growth amid digital disruption & competition

# III. RESULTS & ANALYSIS

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# Results: Churn Model

Table 1 – Result among Machine Learning and Deep Learning

| Model                                  | Accuracy | F1 Score | Precision | Recall | Log Loss | ROC AUC | Training Time (s) |
|----------------------------------------|----------|----------|-----------|--------|----------|---------|-------------------|
| RandomForest                           | 0.82     | 0.82     | 0.82      | 0.82   | 0.40     | 0.90    | 51.77             |
| GradientBoosting                       | 0.87     | 0.87     | 0.88      | 0.86   | 0.31     | 0.95    | 46.01             |
| XGBoost                                | 0.86     | 0.86     | 0.87      | 0.85   | 0.33     | 0.94    | 2.90              |
| AdaBoost                               | 0.77     | 0.76     | 0.80      | 0.73   | 0.52     | 0.85    | 19.46             |
| CatBoost                               | 0.85     | 0.85     | 0.86      | 0.85   | 0.35     | 0.93    | 2.44              |
| LightGBM                               | 0.87     | 0.87     | 0.88      | 0.86   | 0.31     | 0.95    | 2.51              |
| LogisticRegression                     | 0.74     | 0.74     | 0.74      | 0.75   | 0.51     | 0.83    | 0.46              |
| DecisionTree                           | 0.78     | 0.78     | 0.80      | 0.75   | 0.44     | 0.87    | 0.66              |
| Deep Feed-Forward Neural Network (DFF) | 0.80     | 0.80     | 0.81      | 0.79   |          | 0.89    | 5.18              |

# Churn Prediction Model: Gradient Boosting vs. Others

## Gradient Boosting:

- **Accuracy:** 87.4%
- **F1-score:** 87.3% (Precision: 88.4%, Recall: 86.2%)
- **ROC-AUC:** 0.947 → Excellent class separation
- Low false positives & negatives → Crucial for churn management

## Out Performance Most Model

### Model Selection Justification

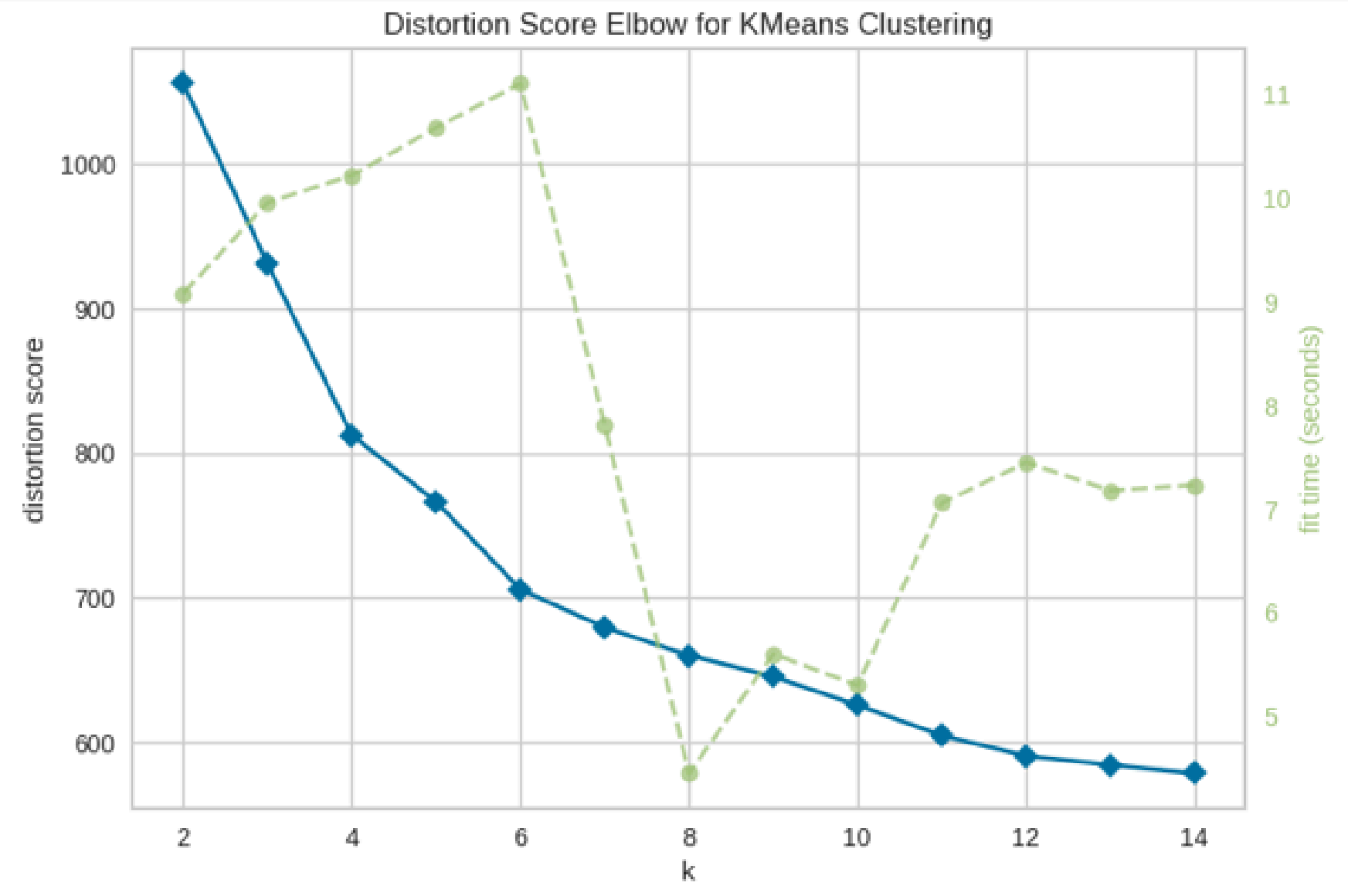
- Gradient Boosting (GB) chosen over:
  - LightGBM: Faster (2.51s), similar accuracy (87%)
  - But less interpretable due to leaf-wise splits
- Interpretability of GB is critical for:
  - Regulatory compliance in banking
  - Auditability and feature transparency



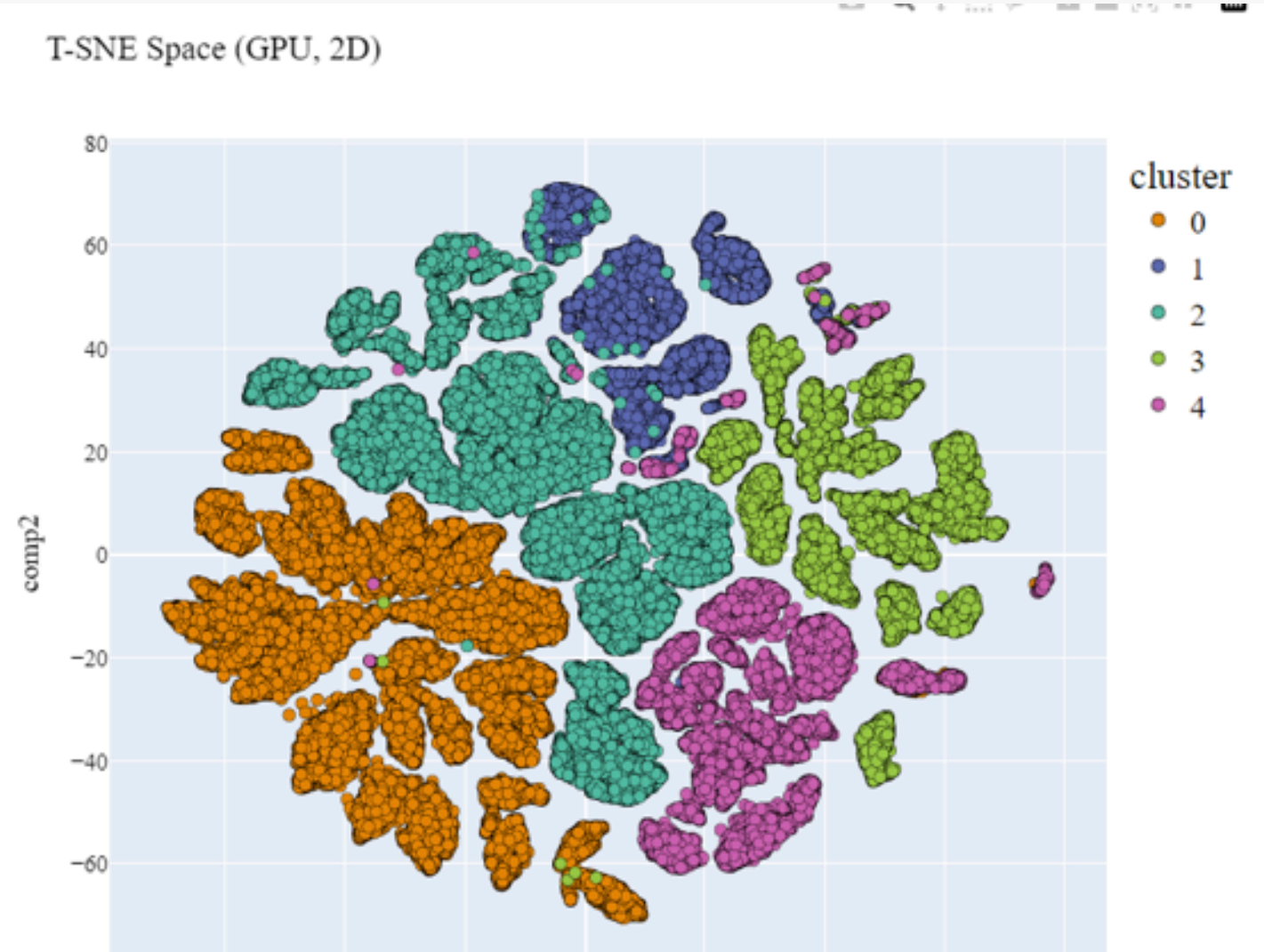
# Clustering Model Results

**Table 2** – Result of KMeans ++ and LLMs Embedding in Train Dataset.

| Inertia | Number of Clusters | Davies-Bouldin Score | Calinski-Harabasz Index |
|---------|--------------------|----------------------|-------------------------|
| 752.56  | 5                  | 1.6072899867         | 62801.38136292665       |



**Fig 3** – Distortion Score Elbow for Kmeans ++ Clustering in Train Dataset.



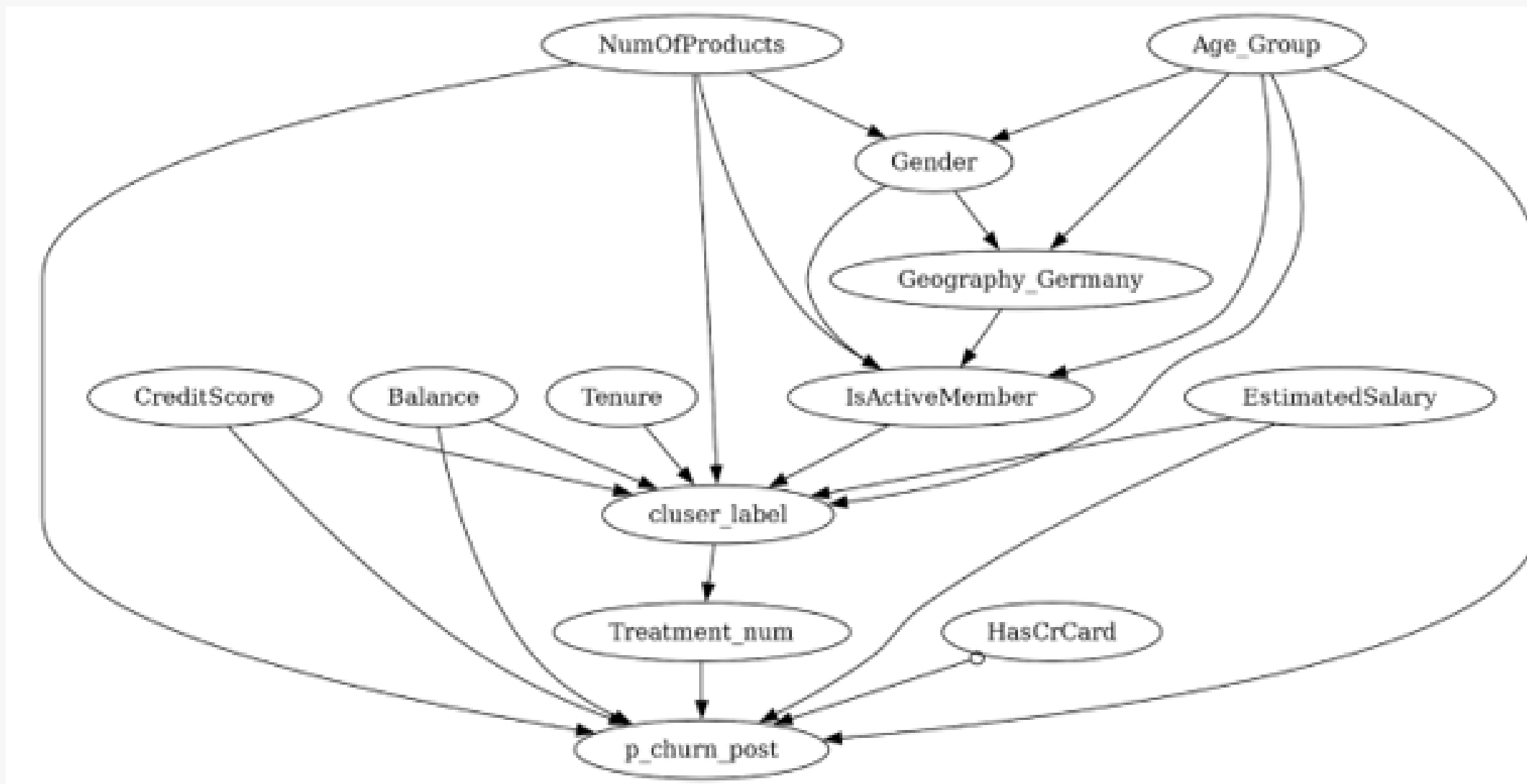
**Fig 4** – T-SNE Space 2D in Train Dataset.

# Five Distinct Clusters & Tailored Strategies

Table 3 – Churn Rate Across Clusters.

| Cluster Characterists                                     | Treatment Strategy         | Churn Rate |
|-----------------------------------------------------------|----------------------------|------------|
| Long-standing customers                                   | Wealth Accumulator Program | 9.89%      |
| Affluent but disengaged customers                         | Engage & Elevate Strategy  | 30.78%     |
| Young customers with lower balances but higher engagement | Starter Growth Plan        | 3.32%      |
| Low account balances and minimal engagement               | Reconnect & Reward Program | 44.48%     |
| Moderate engagement and financial stability               | Premium Balance Rewards    | 18.76%     |

# From Correlation to Causation



**Fig 5** – Final Causal Graph in Total Dataset.

**The Fast Causal Inference (FCI) algorithm was selected**

## Three Level of Causal AI:

### 1. Association

- Seeing / Observing

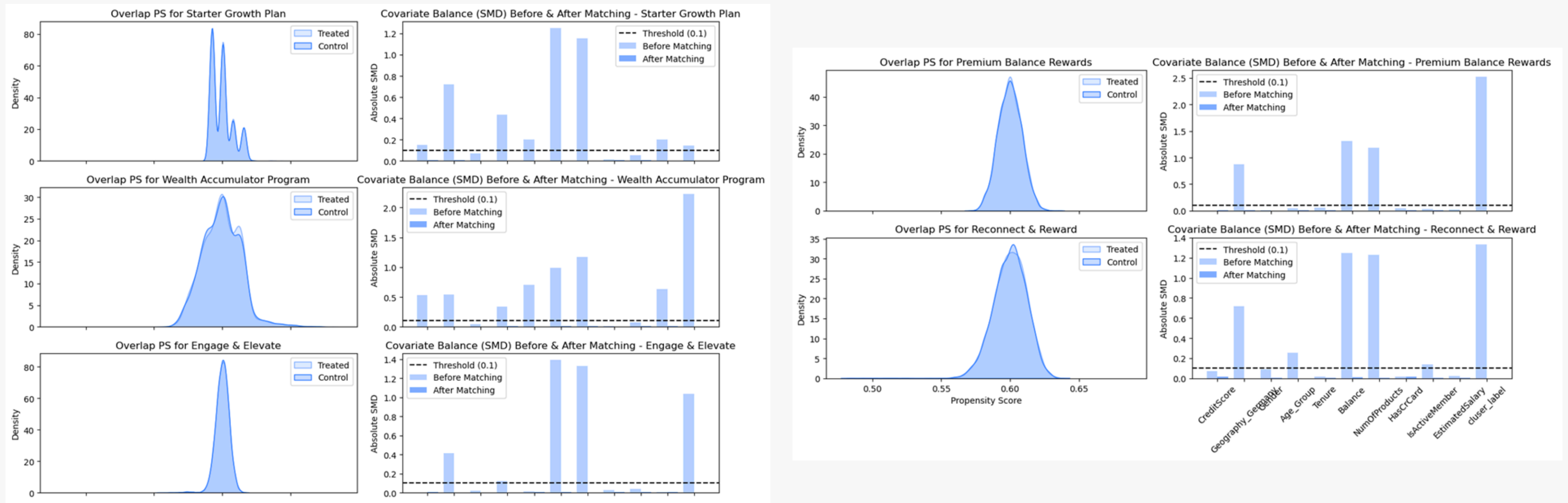
### 2. Intervention

- Addresses the Effect of the Treatment

### 3. Counterfactuals

- Imagining

# Hybrid Propensity Score Matching (HPSM)



**Fig 5** – Overlap PS and SMD for Each Cluster in Total Dataset.

# ATE Using HPSM

**Table 3** – Average Treatment Effect (ATE) of Each Retention Strategy

| Cluster | Treatment Strategy         | Churn Reduction vs Baseline (HPSM) – In Total Data |
|---------|----------------------------|----------------------------------------------------|
| 0       | Wealth Accumulator Program | -4.06%                                             |
| 1       | Engage & Elevate Strategy  | -7.18%                                             |
| 2       | Starter Growth Plan        | -1.72%                                             |
| 3       | Reconnect & Reward Program | -8.24%                                             |
| 4       | Premium Balance Rewards    | -4.79%                                             |



# ATE Using DRLearner

**Table 3** – Average Treatment Effect (ATE) of Each Retention Strategy **in test dataset**

| Cluster | Treatment Strategy         | Churn Reduction vs Baseline (DRLearner) in Test Data | Churn Reduction vs Baseline (DRLearner) in Test Data after Optimization. |
|---------|----------------------------|------------------------------------------------------|--------------------------------------------------------------------------|
| 0       | Wealth Accumulator Program | -16.03%                                              | -27.81%                                                                  |
| 1       | Engage & Elevate Strategy  | -18.83%                                              | -23.05%                                                                  |
| 2       | Starter Growth Plan        | -0.58%                                               | -33.14%                                                                  |
| 3       | Reconnect & Reward Program | -10.73%                                              | -42.39%                                                                  |
| 4       | Premium Balance Rewards    | -17.10%                                              | -30.95%                                                                  |

**Maximum churn reduction: 42.39% in a high-risk cluster.**

# Validating DRLearner

**Table 3** – Average Treatment Effect (ATE) of Each Retention Strategy

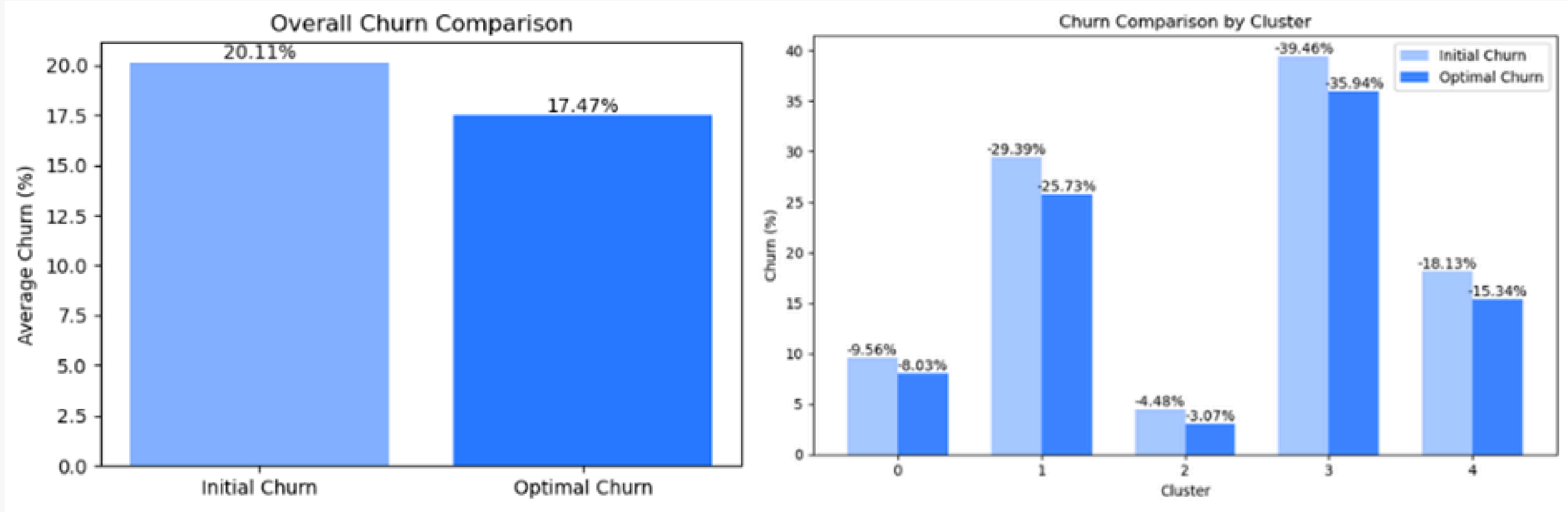
| Refutation Test           | Estimated Effect               | New Effect        |
|---------------------------|--------------------------------|-------------------|
| Random Common Cause       | -0.0155                        | -0.015501         |
| Unobserved Common Cause   | -0.0155                        | (-0.0154, 0.0003) |
| Placebo Treatment         | -0.0155                        | -0.000008         |
| Data Subset Refuter       | -0.0155                        | -0.015456         |
| MAE (Mean Absolute Error) | RMSE (Root Mean Squared Error) | R² (R-squared)    |
| 0.0654                    | 0.1018                         | 0.8362            |

# Examples

**Table 4** – Example of One Customer: Reassigning Treatments and Predicted Churn.

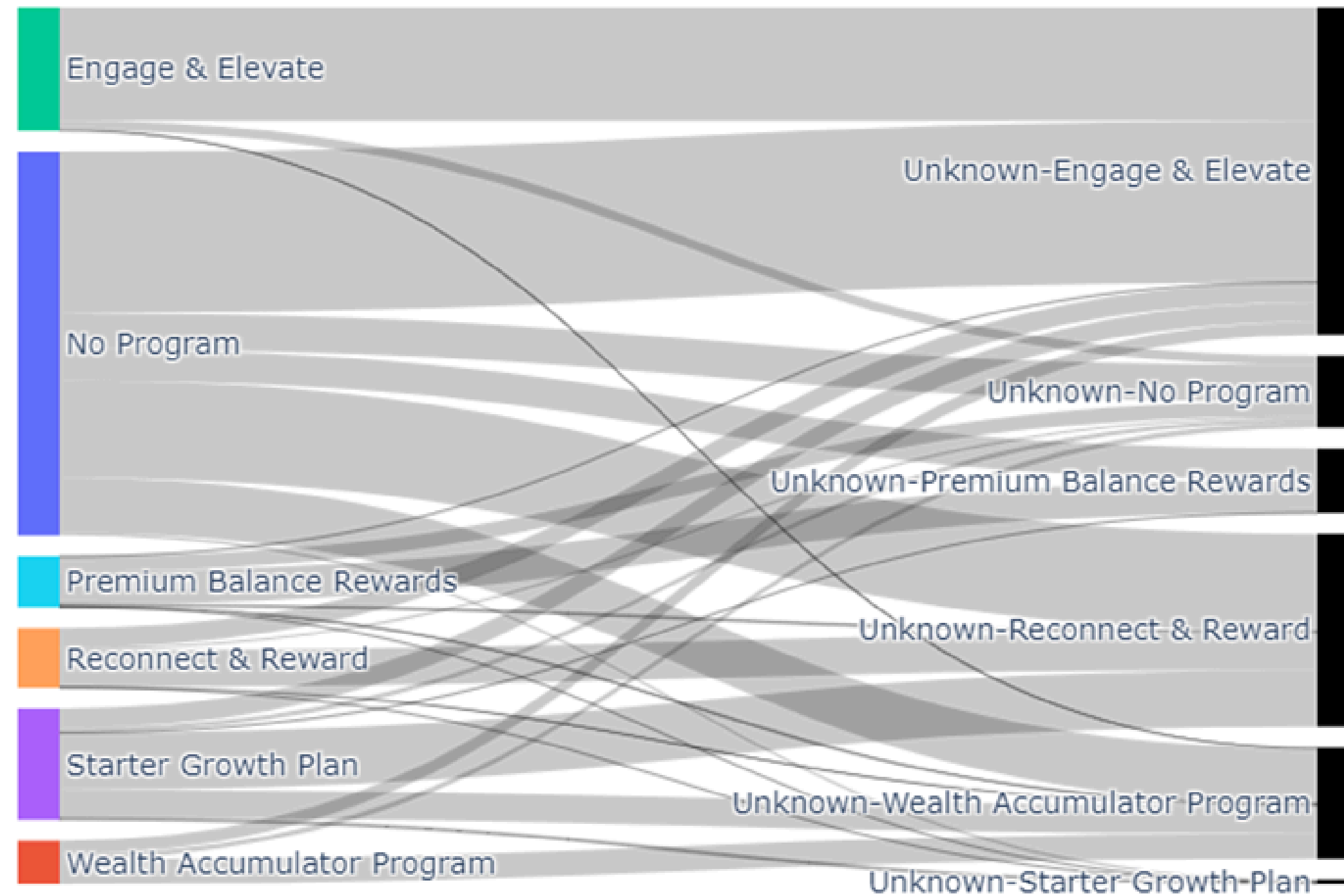
| pred_No<br>Program | pred_Wealth<br>Accumulator | pred_EngageElevate | pred_Star<br>terGrowth | pred_ReconnectReward | pred_PremiumBalance | cluser_label | T_test | treatment_new         | y_test |
|--------------------|----------------------------|--------------------|------------------------|----------------------|---------------------|--------------|--------|-----------------------|--------|
| 0.106109           | 0.147728                   | 0.088637           | 0.13776                | <b>0.324787</b>      | 0.123947            | 3            | 3      | Reconnect<br>& Reward | 0.0934 |

# EXPLAINABLE IN TEST DATA



**Fig 6** – Churn Comparison Before and After Optimization.

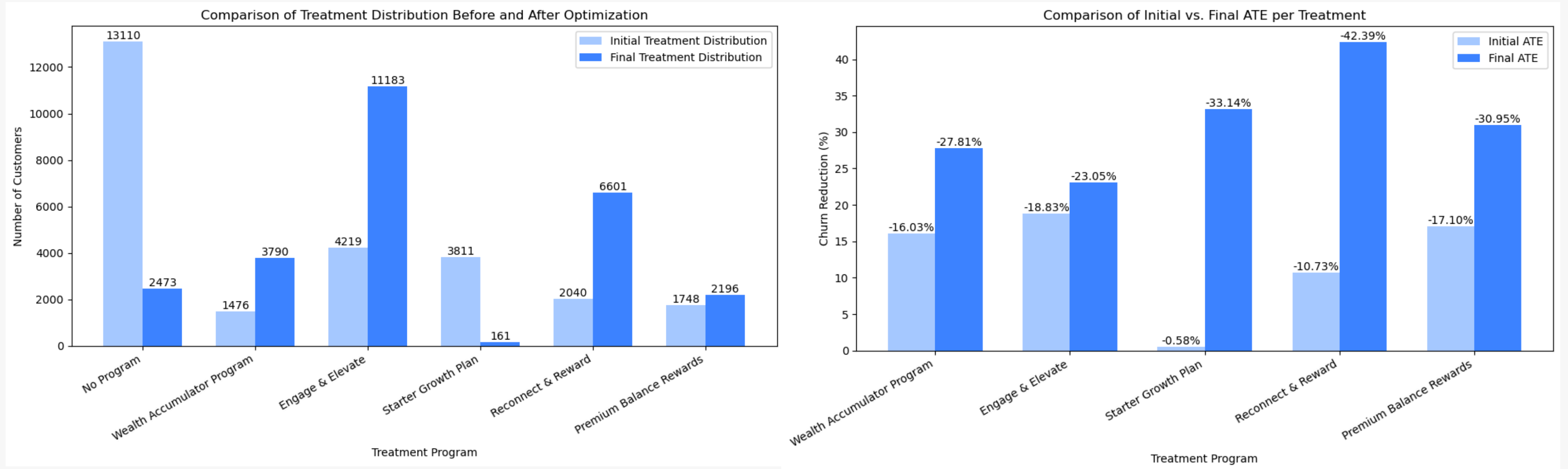
## Treatment Transition Before and After Optimization



**Fig 8** – Snakey Chart show the flow of treatment



# EXPLAINABLE



**Fig 7 – Comparison of Treatment Distribution and Churn Reduction.**

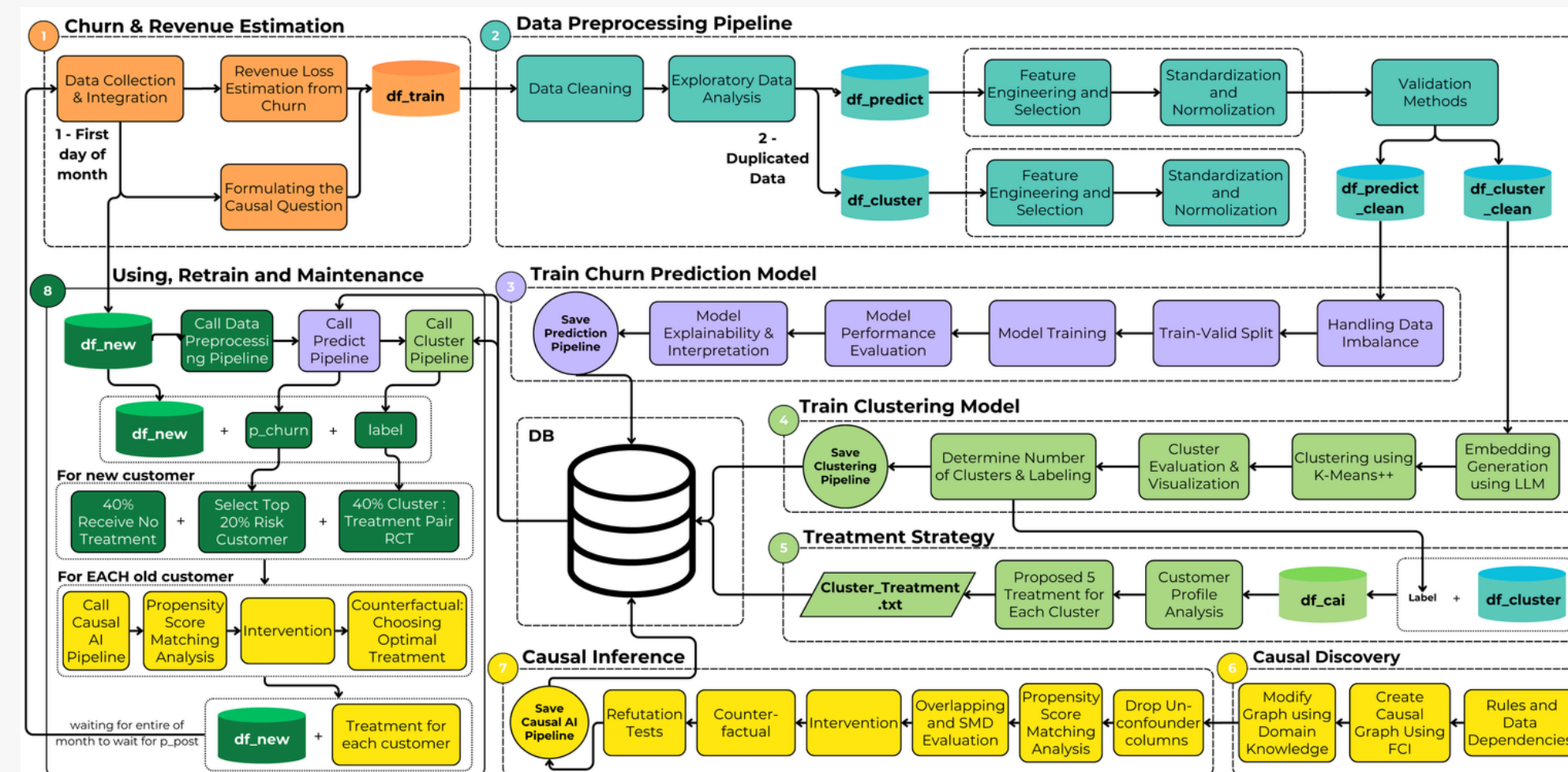
# IV. CONCLUSION & PRACTICAL IMPLICATIONS

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# Contributions

- **Novel Integrated Framework**
  - Combines predictive modeling, LLM-based clustering, and causal inference
- **Segment-Specific Interventions**
  - Tailored strategies for different customer profiles
- **Multi-Treatment Analysis**
  - Evaluates multiple retention actions simultaneously
- **Practical & Actionable**
  - Directly applicable to CRM systems, offering quantifiable ROI



# Limitations & Future Works

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## Limitations

- Observational Data: May contain hidden confounders
- Single-Timeframe Analysis: Doesn't capture evolving customer behavior
- Model Generalizability: Need real-world pilot tests for validation

## Future Works

- Real-Time streaming algorithms      Adaptation: Incorporate data, dynamic bandit
- RCT/Field Experiments: Validate causal effects under controlled conditions
- Enriched Data Sources: Integrate social media, psychographics, or call transcripts
- Explainable AI: Deeper interpretability tools for regulatory

# Conclusion



## **Churn Management: More Than Prediction**

Shift from “Who will churn?” to “Which intervention works, and for whom?”

## **Causal AI**

Identifies true effect of each treatment and each customer

## **High Potential ROI**

Targeted strategies can significantly reduce churn



# Thank You So Much!

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Thesis Defense

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